Ojas Pradhan

4200 Powelton Ave., Unit 2
Philadelphia, PA, USA

pradhanojas23@gmail.com
Linkedin: pradhanojas
+1 (856) 366-0722

Education

Drexel University — Philadelphia, PA

Ph.D., Architectural Engineering — Expected – May 2023
M.Sc., Mechanical Engineering — June 2019
B.Sc., Mechanical Engineering — June 2019

Research Interests

Use of artificial intelligence for sustainable built environment, data-driven fault investigation, smart building control, grid interactive efficient buildings

Research Experience

Graduate Research Assistant — June 2019 - Present
Drexel University — *Philadelphia, PA*

- Contribute to multi-disciplinary projects aimed at developing automated data-driven strategies to detect, diagnose and predict HVAC faults in buildings
- Develop and deploy data predictive supervisory control to optimize energy consumption and recommend control strategies to multi-use building systems
- Collaborate and coordinate with faculty, staff scientists, and fellow graduate students across various universities and research laboratories

Undergraduate Research Assistant — October 2017 - June 2019
Drexel University — *Philadelphia, PA*

- Implemented various whole building HVAC system faults in difference campus buildings to test fault diagnosis algorithms
- Analyzed daily operational data for chiller and air handler units located at demo buildings
- Evaluated fault impact on energy consumption and system operation
- Performed literature review and comparison studies to evaluate existing dynamic building simulation environments
- Developed user and data interface to simulate dynamic hydronic system of a research laboratory using TRNSYS and MATLAB
- Recorded and published simulation results for future deployment of intelligent control algorithms to improve building system operation

Technical Skills

Programming Languages: Python, Matlab, R, SAS, HTML/CSS
Data Science: Machine Learning (Keras/Tensorflow, PyTorch), Deep Learning (CNNs, RNNs), Statistical Inferences, Data Visualization
Database Management: SQL, NoSQL, Spark
Cloud-based Technologies: Microsoft Azure, Docker
Engineering Tools: Dymola, Simulink, TRNSYS, EnergyPlus, LabVIEW, ANSYS
Others: Git, LaTeX

Journal Publications

1. Chen, Y., Wen, J., **Pradhan, O.**, Lo, J., & Wu, T. (2022). Using Discrete Bayesian Networks for Diagnosing and Isolating Cross-level Faults in HVAC Systems. Applied Energy.

2. Huang, J., Yoon, H., Wu, T., Candan, K. S., **Pradhan, O.**, Wen, J., & O'Neill, Z. (2022). Eigen-Entropy: A Metric on Multi-variate Sampling Decision. Information Sciences. (Accepted)

3. Huang, J., Yoon, H., **Pradhan, O.**, Wu, T., Wen, J., O'Neill, Z., & Candan, K. S. (2022). A Cosine-based Correlation Information Entropy Approach for Building Automatic Fault Detection Baseline Construction. Science and Technology for the Built Environment.

4. Huang, J., Wen, J., Yoon, H., **Pradhan, O.**, Wu, T., O'Neill, Z., & Candan, K. S. (2022). Real vs. Simulated: questions on the capability of simulated datasets on building fault detection for energy efficiency from a data-driven perspective. Energy and Buildings.

Conference Presentations

1. **Pradhan, O.**, Hälleberg, D., Chen, Z., Wen, J., Wu, T., Candan, K. S., & O'Neill, Z (2022). Lagged-kNN Based Data Imputation Approach for Multi-Stream Building Systems Data. 7th International High-Performance Buildings Conference at Purdue. July 11-14, 2022.

2. **Pradhan, O.**, Wen, J., Chen, Y., Lu, X., Chu, M., Fu, Y., O'Neill, Z, Wu, T., & Candan, K. S. (2021). Dynamic Bayesian network-based fault diagnosis for ASHRAE guideline 36: high performance sequence of operation for HVAC systems. In Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (pp. 365-368)

3. Huang, J., Wu, T., Yoon, H., **Pradhan, O.**, Wen, J., & O'Neill, Z. (2021). Automatic Fault Detection Baseline Construction for Building HVAC Systems using Joint Entropy and Enthalpy. In IIE Annual Conference Proceedings (pp. 536- 541)

4. **Pradhan, O.**, Wen, J., Chen, Y., Wu, T., & O'Neill, Z. (2021). Dynamic Bayesian Network for Fault Diagnosis. 2021 ASHRAE Virtual Conference. June 28 – June 30, 2021

5. **Pradhan, O.**, Pertzborn, A., Zhang, L., & Wen, J. (2020). Development and Validation of a Simulation Testbed for the Intelligent Building Agents Laboratory (IBAL) using TRNSYS. 2020 ASHRAE Annual Conference. June 29 – July 1, 2020

6. Chen, Y., Wen, J., Chen, T., & **Pradhan, O.** (2018). Bayesian Networks for Whole Building Level Fault Diagnosis and Isolation. 5th International High-Performance Buildings Conference at Purdue. July 9-12, 2018

Professional Experience

Mechanical Engineering Coop April – September 2018
Bristol-Myers Squibb Company *Hopewell, NJ*

- Designed, programed and troubleshooted industrial control systems, such as motion/servo controllers, VFDs, PLCs, HMIs along with SCADA and SQL databases
- Researched on energy markets, plant thermal efficiencies, utility rate structures, commodity pricing and regulatory requirements
- Developed engineering solutions and managed projects involving retrofits, optimization and acquisition of process systems
- Verified project benefits and energy savings through pre/post energy audits
- Maintained and documented equipment/system modifications for all cGMP, vivarium and lab spaces
- Utilized Autodesk application suite to review, update and file drawings, records and documentations for new turn over packages

Honors and Awards

- Linda Latham Scholarship - Presented by American Council for an Energy-Efficient Economy (2022)
- Invited PhD Fellow at University of Nebraska-Lincoln (2022)
- ASHRAE Graduate Student Scholarship – Presented by ASHRAE Philadelphia Chapter (2020)
- ASHRAE Undergraduate Student Scholarship – Presented by ASHRAE Philadelphia Chapter (2019)
- Sustainable Energy Fund Scholarship – Presented by Sustainable Energy Fund (2018)

Professional Associations

- American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), Student Member
- Association for Computing Machinery (ACM), Member
- Energy in Buildings and Communities (EBC): Annex 81 – Data Driven Smart Buildings

References

Jin Wen, PhD.
Professor
Department of Civil, Architectural and Environmental Engineering
Drexel University
jinwen@drexel.edu

Yimin Chen, PhD.
Senior Scientific Engineering Associate
Building Technology & Urban Systems Division
Lawrence Berkeley National Laboratory
yiminchen@lbl.gov

David Hälleberg
Digital Business Developer
Technical Development
Castellum
david.halleberg@castellum.se